

ASSIGNMENT 7

- Removing Impediments -

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User Story:

As a user, I want to interact with a large language model (LLM) through the application’s interface, so that I can get real-time assistance with tasks and generate creative text formats.

Problem: The development team is encountering difficulties integrating the LLM API into the user interface (UI). While the team understands the functionalities of both the LLM and the UI, they lack the specific expertise required to connect them seamlessly. This poses a significant roadblock in completing the user story within the current sprint.

FORCE FIELD ANALYSIS

Goal: Integrate a feature to interact with a large language model (LLM) through the application’s user interface (UI).

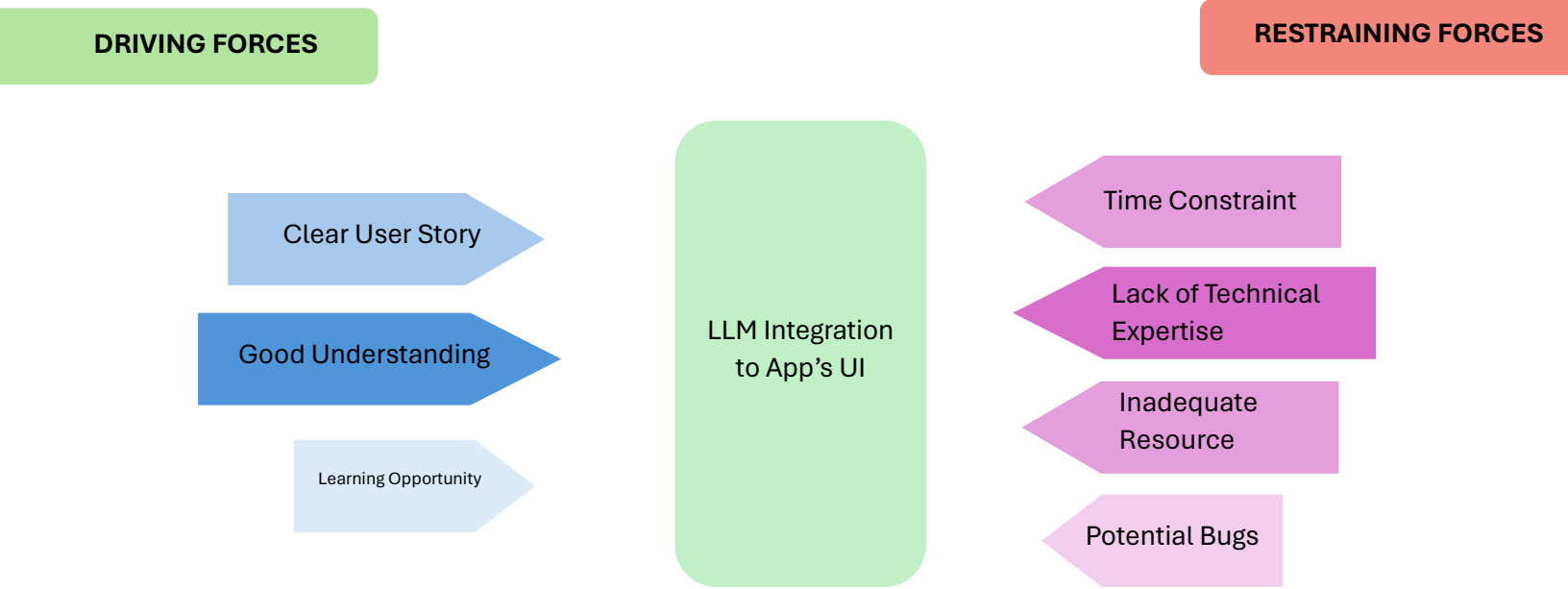
Driving Forces

- **Good Understanding(5):** The team understands the functionalities of both the Large Language Model and the User Interface.
- **Clear User Story(4):** The user story clearly defines the desired goal, making the development direction focused.
- **Learning Opportunity(3):** Integrating the LLM API into the UI will be a valuable learning experience for the team, expanding their skillset.

Restraining Forces

- **Lack of Technical Expertise(5):** The team lacks the specific expertise required to integrate LLM API seamlessly into the application’s UI.
- **Time Constraints(4):** The current sprint has a tight schedule, additional time for learning the specific expertise required in creating the feature could lead to delays.
- **Potential Bugs(3):** Less experience in LLM API integration might pose unforeseen bugs, requiring additional time for fixing.
- **Inadequate Resource(4):** Integrating an API might require cost which may introduce anomaly to sprint’s resource plan.

DRIVING FORCES	RESTRAINING FORCES
Good Understanding – 5 (00%)	Lack of Technical Expertise – 5 (00%)
Clear User Story – 4 (00%)	Time Constraint – 4 (00%)
Learning Opportunity – 3 (00%)	Inadequate Resource – 4 (00%)
	Potential Bugs – 3 (00%)



Analysis and Action

Analyzing these forces made the team realize that even having a lack of expertise being the main roadblock for development, their good understanding on how the LLM and the UI works along with the eagerness to learn its integration give a good driving force. By these, action items the team will take are the following:

- **Research:** Invest a few hours during the sprint to research on how to integrate Large Language Models, how it works within other application’s user interface, as well as the cost of integrating the API.
- **Clarify Requirements:** Create a definitive acceptance criteria that satisfies the integration of LLM API, the primary goal of the sprint, to ensure focus on achieving the requirement.
- **Seek External Help:** Explore external opportunities like consulting developers who have experience integrating LLM APIs or hiring a freelancer for initial implementation.
- **Re-Prioritize:** If research and initial exploration reveal significant roadblocks, consider re-prioritizing the user story for a future sprint when the team has more time or expertise.

Outcome

The incorporation of FFA enabled the team to define the LLM API integration roadblock. More importantly, they identified the actions needed to solve this core problem which significantly aid them to make the informed decision.

They might dedicate time for research and requirement gathering. If the necessary skill is learned, they could proceed in integrating the API along with external help is necessary. Alternatively, they could re-prioritize the user story for a future sprint when the necessary technical skills are acquired as well as a thorough plan is produced, ensuring quality code avoiding bugs as much as possible.